TMUA / PRACTICE PAPER

# Set B Paper 1

Applications of Mathematical Knowledge 20 questions · 75 minutes

## Questions


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*Sequences & Series*

**1.** An infinite geometric series has non-zero first term $a$ and common ratio $r$

The sum to infinity of this series is $S$

The sum to infinity of the squares of the terms of this series is $2S$

The sum to infinity of the cubes of the terms of this series is $\frac{36}{7}S$

What is the value of $S$?

**A** 2

**B** 3

**C** 4

**D** 6

**E** 8

**F** 9

**G** 12

*Algebra*

**2.** A straight line with gradient $m$ passes through the point $(1, 4)$

The line intersects the curve $y = x^2$ at two distinct points with $x$-coordinates $x_1$ and $x_2$

What is the minimum possible value of $x_1^2 + x_2^2$ as $m$ varies over all real numbers?

**A** 1

**B** 4

**C** 7

**D** 8

**E** 12

**F** 16

*Trigonometry*

**3.** How many distinct real values of $x$ satisfy the equation

$$\cos(2x) = 1 - x$$

in the range $0 \leq x \leq \pi$?

**A** 0

**B** 1

**C** 2

**D** 3

**E** 4

**F** 5

**G** 6

*Integration*

**4.** Use the trapezium rule with 4 equal strips to estimate

$$\int_1^5 \ln x \, \mathrm{d}x$$

**A** $\frac{1}{2}\ln(120)$

**B** $\frac{1}{2}\ln(180)$

**C** $\frac{1}{2}\ln(240)$

**D** $\frac{1}{2}\ln(720)$

**E** $\frac{1}{2}\ln(2880)$

**F** $\frac{1}{2}\ln(14400)$

*Coordinate Geometry*

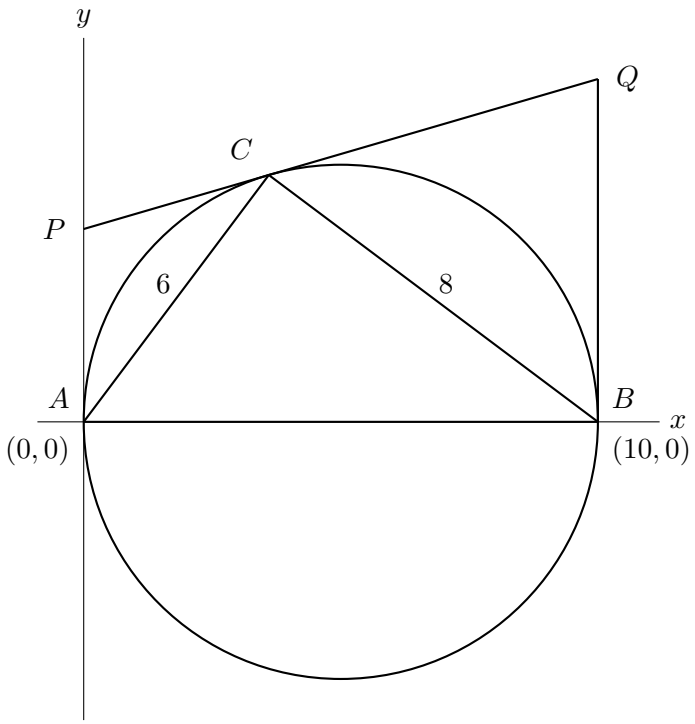
**5.** The points $A$ and $B$ have coordinates $(0, 0)$ and $(10, 0)$ respectively.

A circle has diameter $AB$.

The point $C$ lies on the circle above the $x$-axis, with $AC = 6$ and $BC = 8$.

The tangent to the circle at $C$ meets the lines $x = 0$ and $x = 10$ at $P$ and $Q$ respectively.

What is the length $PQ$?




**A** $\frac{35}{12}$

**B** $\frac{15}{4}$

**C** $\frac{20}{3}$

**D** $10$

**E** $\frac{125}{12}$

**F** $\frac{125}{6}$

*Integration*

**6.** Let $k$ be a positive real constant. The function $f(x)$ is defined for $x > 0$ by

$$f(x) = \int_1^x \frac{t-k}{\sqrt{t}}\,\mathrm{d}t$$

What is the complete set of values of $k$ for which the equation $f(x) = 0$ has exactly two distinct real solutions for $x > 0$?

**A** $k > 1$

**B** $k > \frac{1}{3}$

**C** $k > \frac{1}{3}$ and $k \neq 1$

**D** $\frac{1}{3} < k < 1$

**E** $k > 0$ and $k \neq 1$

**F** $k \geq \frac{1}{4}$ and $k \neq 1$

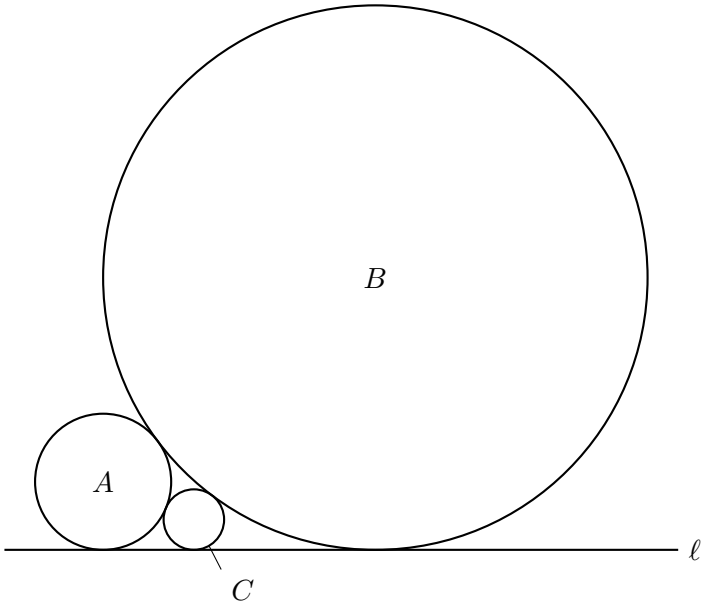
*Geometry*

**7.** A straight line $\ell$ is drawn, and three circles lie on the same side of $\ell$. Each circle touches $\ell$ at exactly one point.

Circle $A$ has radius 1 and circle $B$ has radius 4, and these two circles touch each other.

Circle $C$ lies in the space between circle $A$, circle $B$ and the line $\ell$, touching circle $A$, circle $B$ and $\ell$.

What is the radius of circle $C$?




**A** $\frac{2}{9}$

**B** $\frac{4}{9}$

**C** $\frac{1}{2}$

**D** $\frac{16}{25}$

**E** $\frac{2}{3}$

**F** $\frac{4}{5}$

**G** $\frac{8}{9}$

*Probability & Counting*

**8.** $N = 2^7 \times 3^5$

One of the positive factors of $N$ that is divisible by 12 is selected at random, with each such factor equally likely to be selected.

What is the probability that $N$ divided by the selected factor is a perfect square?

**A** $\frac{1}{6}$

**B** $\frac{1}{5}$

**C** $\frac{1}{4}$

**D** $\frac{3}{10}$

**E** $\frac{1}{3}$

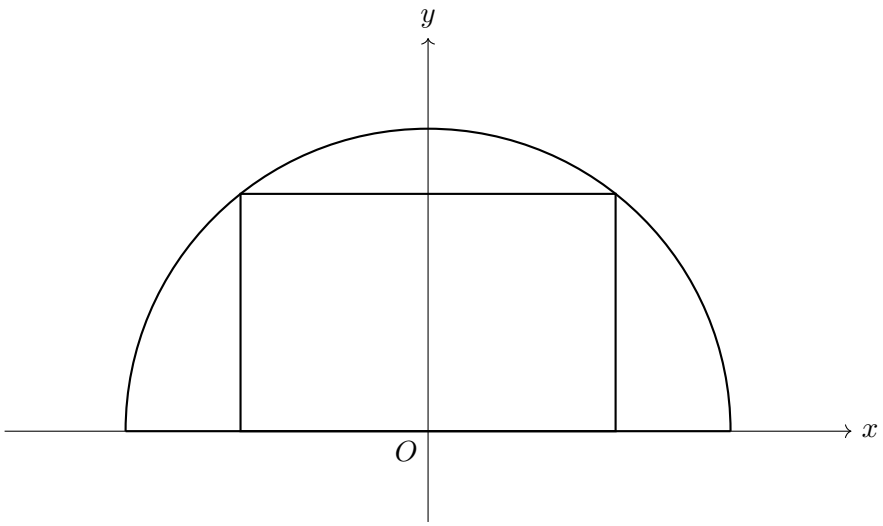
**F** $\frac{2}{5}$

*Geometry*

**9.** The diagram shows a semicircle of radius 1 with its diameter along the $x$-axis and its centre at the origin.

A rectangle has one side on the diameter and its two remaining vertices on the arc of the semicircle.

What is the largest possible perimeter of such a rectangle?




**A** $\sqrt{5}$

**B** $4$

**C** $2+\sqrt{5}$

**D** $3\sqrt{2}$

**E** $2\sqrt{5}$

**F** $5$

*Polynomials*

**10.** Find the complete set of real values of $k$ for which the quadratic equation

$$x^2 - (k+2)x + (2k+1) = 0$$

has two distinct **positive** real roots.

**A** $k > 4$

**B** $k < 0$ or $k > 4$

**C** $-\frac{1}{2} < k < 0$ or $k > 4$

**D** $-2 < k < 0$ or $k > 4$

**E** $k > -\frac{1}{2}$

*Sequences & Series*

**11.** A sequence $a_n$ is defined by $a_1 = 1$ and, for $n \geq 1$,

$$a_{n+1} = a_n + \frac{1}{a_n}$$

Which of the following statements is true for all $n \geq 1$?

**A** $a_n$ is an integer

**B** $a_n^2$ is an integer

**C** $a_n < n$

**D** $a_n^2 \geq 2n - 1$

**E** $a_n^2 \leq 2n - 1$

**F** $a_n \leq \sqrt{2n}$

*Number*

**12.** The positive real numbers $p \times 10^{-4}$, $q \times 10^{-2}$ and $r \times 10^{-1}$ are each in standard form, and

$$(p \times 10^{-4}) + (q \times 10^{-2}) = (r \times 10^{-1})$$

*How many* of the following statements must be true?

**I:** $q > 1$
**II:** $r > 1$
**III:** $q < r$
**IV:** $p < r$

**A** 1 of them

**B** 2 of them

**C** 3 of them

**D** 4 of them

**E** None of them

*Probability & Counting*

**13.** A bag contains only red balls and blue balls, with at least one ball of each colour. The total number of balls is $n$, where $30 < n < 50$.

Two balls are drawn at random without replacement. The probability that they have the same colour is equal to the probability that they have different colours.

How many different possible values are there for the number of red balls in the bag?

**A** 1

**B** 2

**C** 3

**D** 4

**E** 5

*Functions & Graphs*

**14.** For each real value of $x$, let $M(x)$ be the larger of the two numbers

$$(x+3)^2+1 \quad \text{and} \quad 4(x-1)^2+1$$

For which value of $x$ is $M(x)$ least?

**A** $-3$

**B** $-1$

**C** $-\frac{1}{3}$

**D** $\frac{1}{3}$

**E** $1$

**F** $5$

*Number*

**15.** Let $N = 2^6 \times 3^4 \times 5^3$

A positive factor $d$ of $N$ is chosen such that the highest common factor of $d$ and $\frac{N}{d}$ is 60

How many possible values of $d$ are there?

**A** 2

**B** 4

**C** 6

**D** 8

**E** 12

*Polynomials*

**16.** Consider

$$P(x) = x^4 + ax^3 + bx^2 + ax + 1$$

where $a$ and $b$ are real constants.

Suppose $(x - 2)$ and $\left(x + \frac{1}{2}\right)$ are both factors of $P(x)$

What is the value of $b$?

**A** $-13$

**B** $-5$

**C** $0$

**D** $5$

**E** $13$

**F** $-\frac{17}{4}$

*Integration*

**17.** Let $k$ be a positive constant. The curve $C$ has equation

$$y = x^3 - 3k^2x + 2k^3$$

The point $A$ is the local maximum of $C$

The line parallel to the $x$-axis passing through $A$ intersects $C$ again at the point $B$

The area of the region bounded by $C$ and the line segment $AB$ is 108

What is the value of $k$?

**A** 1

**B** $\sqrt{2}$

**C** $\sqrt{3}$

**D** 2

**E** $2\sqrt{2}$

**F** 3

**G** 4

*Algebra*

**18.** Let $a$ and $b$ be real numbers with $a + b = 1$

What is the least possible value of

$$(a^2 + b)(b^2 + a) \quad ?$$

**A** $\frac{9}{16}$

**B** $\frac{1}{4}$

**C** $\frac{1}{2}$

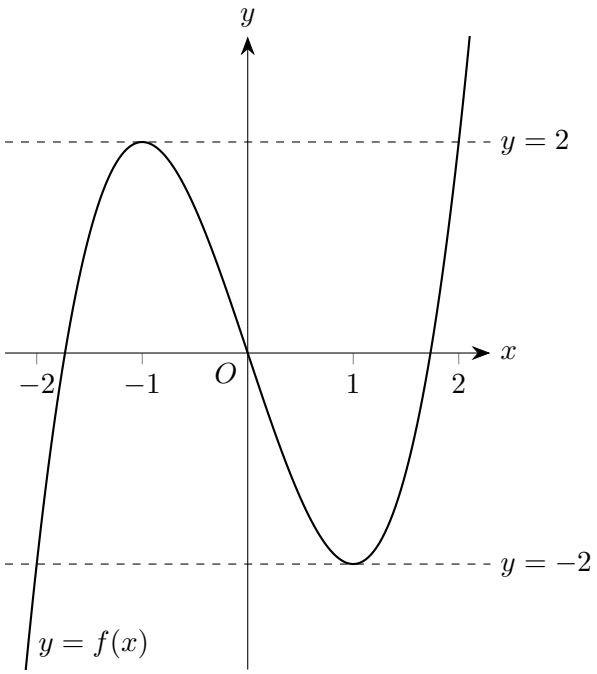
**D** $\frac{3}{4}$

**E** $1$

*Functions & Graphs*

**19.** The diagram shows the graph of $y = f(x)$, where $f(x) = x^3 - 3x$. The horizontal lines through the turning points are also shown.

How many distinct real solutions does the equation $f(f(x)) = f(x)$ have?




**A** 5

**B** 6

**C** 7

**D** 8

**E** 9

*Inequalities*

**20.** Let $f(x) = ax^2 + bx + c$ be a quadratic function with real coefficients such that $f(x) \geq 0$ for all real $x$

Suppose that for every real $x$,

$$f(x+1) - f(x) \geq 2x + 3$$

What is the smallest possible value of $f(0)$?

**A** $0$

**B** $\frac{1}{4}$

**C** $1$

**D** $2$

**E** $\frac{9}{4}$

**F** $4$

**G** $\frac{25}{4}$